2. Read "The Shifting Reasons for Beveridge Curve Shifts" by Barlevy et. al (*Journal of Economic Perspectives*, 2024) with a focus on pages 94–97. How do the authors explain the shifts of the Beveridge curve during the Covid recession and recovery period (represented by the blue data points in your figure) and the unusual shape during the monetary tightening phase (represented by the black data points in your figure)? What is meant by the "Great Resignation"? How and why was the recovery from the pandemic recession different from past recoveries in terms of the Beveridge curve?

OPTIONAL EXERCISES

Exercise 4: Short Questions

Provide a **brief** answer to each question. You should aim for 4-5 sentences per item.

1. Define calibration. Describe its purpose and underlying logic, outlining the main steps involved in calibrating a model to suitably reflect real-world data.

2. What is the Lucas' critique, and what was its primary impact on macroeconomic theory in the 1970s?

3. The employment growth rate from t to t+1 can be expressed either as hires minus separations, or as job creation minus job destruction (all rates measured at t). Explain the conceptual difference between *worker flows* and *job flows*, and discuss their accounting relationship (including replacement hiring). Briefly summarize their distinct cyclical patterns in the JOLTS and BED evidence.

Exercise 5: Durable Consumption

Consider an agent who is planning her intertemporal consumption of a durable good, like housing or cars. The agent derives utility from the stock of the durable good D_t. The problem of the agent is:

$$\max_{\{D_t, a_{t+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \, u(D_t)$$

$$\text{s.t.} \begin{cases} D_t + a_{t+1} = (1+r)\, a_t + (1-\delta)\, D_{t-1} + y_t \\ D_{-1}\,, a_0 \text{ given} \end{cases}$$

where a_t denotes the amount of 1-period bonds held at the beginning of period t, yielding a constant rate of return r. The existing stock of durable goods depreciates at rate δ.

Income evolves stochastically according to the following AR(1) process

$$y_{t+1} = \rho\, y_t + \varepsilon_{t+1}$$

where $\rho \in [0, 1)$, and ε are iid zero-mean shocks.

The utility function is differentiable, strictly increasing, strictly concave, and satisfies the Inada conditions.

1. Formulate the problem recursively: identify the exogenous state variable, the endogenous state variables, and the control variables of the problem, and write down the Bellman equation.

   [HINT: there are 2 endogenous state variables and 2 control variables]

2. Derive the first-order conditions and the two envelope conditions from the recursive formulation. Give an intuitive interpretation of the two Euler equations you obtain (one for bond investment, the other for durable investment); what is the tradeoff between the two forms of investment?

   [HINT: you will have Lagrange multipliers appearing in the two Euler equations; recall that the Lagrange multiplier associated to the resource constraint, λ_t, can be interpreted as the (current) marginal value of resources to the agent (and similarly for λ_{t+1}).]

3. Exploit the two Euler equations to show that the problem yields a standard Euler equation for the *stock* of durables

$$u'(D_t) = \beta\,(1+r)\,\mathbb{E}_t\left[u'(D_{t+1})\right]$$

   [HINT: combine the two Euler equations from the previous point to first show that $\lambda_t = \frac{1+r}{r+\delta}\,u'(D_t)$, which similarly applies for λ_{t+1}. Then, substitute this result into the Euler equation for bonds to eventually get to the required result.]

4. Assume that $u(D_t) = \ln(D_t)$, and log-linearise the Euler equation you obtained in the previous point (be careful when dealing with the expectation operator) to show that the expected (approximate) growth rate of the durable stock is equal to zero.

   [HINT: letting $\widetilde{d}_t := \ln\left(\frac{D_t}{D}\right)$, notice that $\widetilde{d}_{t+1} - \widetilde{d}_t = \ln\left(\frac{D_{t+1}}{D_t}\right) \approx \frac{D_{t+1}-D_t}{D_t}$.]